

4. BIAS VARIANCE TRADE - OFF

Overfitting

Model fitting the training data very well but performing badly on new data.

Overfitting is explained through BVT.

We assume :

$$\begin{array}{ccccc} Y & = & f(X) & + & \epsilon \\ \downarrow & & \downarrow & & \downarrow \\ \text{Actual} & & \text{Input/} & & \text{Random Noise} \\ \text{Outcome} & & \text{Features} & & \end{array}$$

We do not know $f(X)$.
So we estimate a model :

$$\hat{f}(X)$$

Our prediction is :

$$\hat{Y} = \hat{f}(X)$$

The prediction error is

$$Y - \hat{f}(X)$$

So, we need to choose $\hat{f}$ so that prediction error is low on new unseen data and not just on training data.

$$\text{Training Error} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

$$\text{Test Error} = \mathbb{E} [(Y_{\text{new}} - \hat{f}(X_{\text{new}}))^2]$$

Overfitting $\rightarrow$ Model memorised the Training Data.

$$\text{Expected Test Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

$$\mathbb{E} [(Y - \hat{f}(x))^2] = \text{Bias} [\hat{f}(x)]^2 + \text{Var} [\hat{f}(x)] + \sigma^2$$

$$\text{where, } \sigma^2 = \text{Var}(\epsilon)$$

Bias

→ How far the avg. prediction is from the true function.

$$\text{Bias}[\hat{f}(x)] = \mathbb{E}[\hat{f}(x)] - f(x)$$

$$\text{Bias}[\hat{f}(x)]^2 = (\mathbb{E}[\hat{f}(x)] - f(x))^2$$

A high Bias model makes strong simplifying assumptions.

E.g. If the true relationship is curved, but we fit a straight line.

$$Y = X^2 + \epsilon \quad (\text{Curved})$$

$$\hat{Y} = \beta_0 + \beta_1 X \quad (\text{Straight Line})$$

High Bias models usually **underfit**. E.g.

- Using only one explanatory variable when many matter.
- Use too much regularisation

Variance

→ Measures how much the fitted model changes when the training data changes.

$$\text{Var}[\hat{f}(x)] = \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]$$

A high variance model reacts too strongly to noise in the training data. It does not just learn the signal, it learns random fluctuations.

High Variance models usually **overfit**. They are too flexible. E.g.

- Very high degree polynomial regression
- Decision Trees with no depth limit.
- Nearest Neighbour models with very small k .
- NNs trained too long on small data
- Models with too many predictors relative to observation.
- Weak Regularisation

Very low Trng. Error, but high Test Error.

Irreducible Error

→ Noise that no model can eliminate.

$$\sigma^2 = \text{Var}(\epsilon)$$

Sudden Geopolit. Events

Market Panic

Liquidity Shocks

Data Revisions

Measurement errors in GDP etc.

So, the lowest possible prediction error is σ^2 and not zero.

As model complexity increases, Bias Decreases & Variance Increases.

As model " decreases, Bias Increases & Variance Decreases

Since, Irreducible Error is cannot be controlled, the practical goal is

$$\min(\text{Bias}^2 + \text{Variance})$$

As model complexity increases, Training Error usually falls continuously as a more complex model can fit the training data better.

But Test Error usually behaves like a U-shape. At first increasing complexity helps, but after a point the model starts learning noise.

OLS Linear Regression can be High Bias or High Variance depending on the situation. If the true relationship is linear and assumptions are reasonable, OLS can work very well.

But if the true relationship is non-linear, OLS may have high Bias.

If we add too many polynomial terms, OLS can become high variance. e.g.

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_{50} X^{50} + \epsilon$$

So, within Linear Regression, the BVT appears through :

- # Predictors
- Interaction Terms
- Sample Size
- Polynom. Degree
- Regularisation Strength

Financial Markets have low Signal-to-Noise ratio. So, high Variance models can easily discover fake patterns.

In finance, Overfitting is especially dangerous because historical return patterns can be noisy, unstable and regime dependent.

In our Problem 3, Temperature Bins reduce bias because they allow flexible non-linear effects.

But too many of those bins can increase variance, esp. if some temp. ranges have few observations.

Mathematical Derivation of the Bias-Variance Decomposition

Let true data generating process be :

$$Y = f(x) + \epsilon$$

where,

$$\begin{aligned} E[\epsilon] &= 0 \\ \text{Var}(\epsilon) &= \sigma^2 \end{aligned}$$

We estimate $\hat{f}(x)$

The expected squared prediction error is :

$$E[(Y - \hat{f}(x))^2]$$

Substitute $Y = f(x) + \epsilon$

So,

$$\begin{aligned} &E[(f(x) + \epsilon - \hat{f}(x))^2] \\ &= E[(f(x) - \hat{f}(x) + \epsilon)^2] \end{aligned}$$

Expanding,

$$E[(f(x) - \hat{f}(x))^2] + E[\epsilon^2] + 2E[\epsilon(f(x) - \hat{f}(x))]$$

Since, Noise has mean of zero and is independent of the model estimate

So, $E[\epsilon(f(x) - \hat{f}(x))] = 0$

So,

$$E[(Y - \hat{f}(x))^2] = E[(f(x) - \hat{f}(x))^2] + E[\epsilon^2]$$

$$\mathbb{E}[(Y - \hat{f}(x))^2] = \mathbb{E}[(f(x) - \hat{f}(x))^2] + \sigma^2$$

We decompose, $\mathbb{E}[(f(x) - \hat{f}(x))^2]$

$$\Rightarrow f(x) - \hat{f}(x) = \underbrace{f(x) - \mathbb{E}[\hat{f}(x)]}_A + \underbrace{\mathbb{E}[\hat{f}(x)] - \hat{f}(x)}_B$$

$$f(x) - \hat{f}(x) = A + B$$

$$\begin{aligned}\mathbb{E}[(f(x) - \hat{f}(x))^2] &= \mathbb{E}[(A+B)^2] = \mathbb{E}[A^2 + B^2 + 2AB] \\ &= \mathbb{E}[A^2] + \mathbb{E}[B^2] + 2\mathbb{E}[AB]\end{aligned}$$

$$A = f(x) - \mathbb{E}[\hat{f}(x)]$$

This is not random once x is fixed. So,

$$\mathbb{E}[A^2] = A^2$$

$$\mathbb{E}[A^2] = \underbrace{(f(x) - \mathbb{E}[\hat{f}(x)])^2}_{\text{Squared Bias}}$$

It can alternatively be written as :

$$\text{Bias}[\hat{f}(x)]^2 = (\mathbb{E}[\hat{f}(x)] - f(x))^2$$

$$B = \mathbb{E}[\hat{f}(x)] - \hat{f}(x)$$

$$\mathbb{E}[B^2] = \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2]$$

$$\text{Var}[\hat{f}(x)] = \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]$$

$$2\mathbb{E}[AB] : 2\mathbb{E}[\underbrace{(f(x) - \mathbb{E}[\hat{f}(x)]) \cdot (\mathbb{E}[\hat{f}(x)] - \hat{f}(x))}_{\text{Not random w.r.t. true sample}}]$$

Taking it outside the expectation

$$\begin{aligned}
 &= 2(f(x) - \mathbb{E}[\hat{f}(x)]) \cdot \mathbb{E}[\mathbb{E}[\hat{f}(x)] - \hat{f}(x)] \\
 &= 2(f(x) - \mathbb{E}[\hat{f}(x)]) \cdot (\mathbb{E}[\hat{f}(x)] - \mathbb{E}[\hat{f}(x)])
 \end{aligned}$$

$$2 \mathbb{E}[AB] = 0$$

$$\begin{aligned}
 \mathbb{E}[(f(x) - \hat{f}(x))^2] &= \mathbb{E}[A^2] + \mathbb{E}[B^2] + 0 \\
 &= \text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)]
 \end{aligned}$$

So,

$$\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

$$\text{Prediction Error} = \text{Systematic Model Error} + \text{Model Instability} + \text{Unavoidable Noise}$$

$\mathbb{E}[\hat{f}(x)] \rightarrow$ Avg. prediction we would get if we repeatedly trained the model on many different training samples drawn from the same population.

Recall the Darts game analogy for Bias & Variance.

Goal is to find the complexity level where test error is minimised.

Regularisation $\rightarrow$ A tool for managing the BVT.
It adds a penalty to model complexity.

For Ridge Regression :

$$\min_{\beta} \left[\text{RSS} + \lambda \sum_{j=1}^p \beta_j^2 \right]$$

For LASSO :

$$\min_{\beta} \left[\text{RSS} + \lambda \sum_{j=1}^p |\beta_j| \right]$$

When λ increases, coefficients shrink. That usually increases Bias but reduces Variance.

A little regularisation can improve test performance. Too much regularisation causes underfitting.

Cross Validation $\rightarrow$ In practice we do not directly observe the true Bias & Variance

So, we use Validation or Cross Validation

On **k-fold Cross Validation** —

- $\hookrightarrow$ Split data into k-parts
- $\hookrightarrow$ Train on k-1 parts
- $\hookrightarrow$ Validate on remaining part
- $\hookrightarrow$ Repeat k times
- $\hookrightarrow$ Avg. Validn. error

CV Score is :

$$CV = \frac{1}{k} \sum_{j=1}^k \text{Error}_j$$

We choose the model complexity or hyperparameter that gives the lowest Validation Error.

Cross Validn. is the practical tool for finding the bias-variance sweet spot.

Bias-Variance in KNN $\rightarrow$ KNN is based on the avg. of nearby observations.

If $k=1$, model uses only the closest observn. This is very flexible and has Low Bias and High Variance.

If $k=100$, model averages many neighbours. This is smoother and has High Bias and Low Variance.

Bias-Variance in Decision Trees $\rightarrow$ A deep tree can split

the data many times, and may fit training data almost perfectly but may generalise poorly $\rightarrow$ High Var.

A shallow tree — more stable but may miss important patterns — High Bias.

Random Forests reduce Variance by averaging many trees.
Boosting often reduces bias by sequentially improving weak learners.

Small Data → Prefer Simpler Models

Large Data → Complex models become safer

As sample size increases, variance usually falls.

In finance — long historical time series may still contain few independent regimes.

Adding more features can reduce Bias — but adding irrelevant features can increase Variance and the model may overfit.

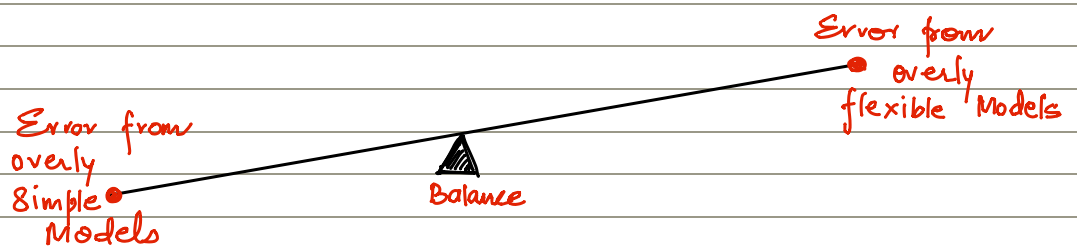
So Feature Engineering must balance

Information
Gain

against

Variance
Increase

Model choice is not only about accuracy. It also depends on interpretability, stability, sample size and the cost of errors.



Regularised Regression General Form :

$$\text{Loss} = \text{Training Error} + \text{Penalty}$$

Good ML is about finding the model complexity that minimises the Test Error rather than the Training Error.